
Scatter Plot Practice Tasks

☐ Task 1 — Size Based on Data

Create:

- `study_hours`
- `marks`

Set:

```
sizes = marks * 5
```

☞ Higher marks = bigger points.

☐ Task 2 — Conditional Colors

Apply colors using conditions:

- `marks < 40` → red
- `marks 40–70` → orange
- `marks > 70` → green

Hint:

```
colors = np.where(marks>70, 'green', np.where(marks>40, 'orange', 'red'))
```

☐ Task 3 — Alpha Comparison

Create 3 scatter plots:

- `alpha = 1`
- `alpha = 0.5`
- `alpha = 0.2`

☞ Observe overlapping effects.

□ Task 4 — Employee Attrition Visualization (ML Style)

Use employee-like data:

- x = salary
 - y = performance_score
 - color = attrition (0 blue, 1 red)
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□ Task 5 — Three Groups Scatter

Create 3 groups:

- Beginners
- Intermediate
- Advanced

Each group should have:

- different color
 - different size
 - label + legend title
-

□ Task 6 — Add Mean Lines

Add average lines:

```
plt.axhline(mean_y)
plt.axvline(mean_x)
```

☞ Makes plots look professional.

□ Task 7 — Bubble Chart

Instead of random size:

```
revenue = np.random.randint(1000, 5000, size=50)
sizes = revenue/10
```